

Contents

Morpheus: On-Device Predictive Autocompletion for Basque Using State Space

Models	1
Abstract	1
1. Introduction	2
2. Related Work	3
3. Architecture Selection	5
4. Model Architecture & Training	7
5. Deployment Pipeline	14
6. Evaluation	17
7. Future Work	30
8. Code Availability	32
9. Conclusion	33
Appendix A: The Morfessor Attempt and Pivot	35
Appendix B: Where 4K Still Fails — Full Failure-Mode Table	35
Appendix C: Inference Engine Sensitivity — The SSM_SCAN Bug	36
Appendix D: CSR Inversion — Full GGUF Cross-Validation Table	36
References	36

Morpheus: On-Device Predictive Autocompletion for Basque Using State Space Models

Preprint — July 2026

Authors: Xabier Ezpeleta

Code & Models: github.com/itzune/morpheus-mamba

Live Demo: Docker-based local deployment supporting both Smart Compose-style ghost text and smartphone keyboard-style word chips

Keywords: predictive autocompletion, Basque, Euskara, Mamba, State Space Models, agglutinative languages, on-device inference, keystroke savings, evaluation methodology, next-word prediction, subword tokenization

Abstract

We present **Morpheus**, an on-device predictive autocompletion system for Basque (Euskara), a low-resource agglutinative language. The research question is whether a Gmail Smart Compose-equivalent multi-token continuation system — which Google serves from Cloud TPUs using an ~80M-parameter LSTM trained on ~8 billion emails — can run **entirely on a consumer laptop**, with zero network calls, for use in text editors. We select Mamba-2, a Selective State Space Model offering constant-memory $O(1)$ per-step inference with no KV cache — the same property (no KV cache) that led Google to choose LSTM over Transformer for Smart Compose, but achieved here with a modern architecture that runs on-device.

Our model (91M parameters, 55 MB Q4_K_M) is trained on ~10B tokens of curated Basque text using a 4K-vocabulary SentencePiece Unigram tokenizer. **A controlled vocabulary-size**

ablation across 4K, 8K, 16K, and 32K reveals that MorphAcc consistency drops from 66.7% at 4K to 28.6% at 32K, mirroring the QuechuaTok finding. We argue that **fertility is a confound, not a quality metric**, for agglutinative tokenizer evaluation: lower fertility is achieved exactly by fusing morphemes into opaque units.

After full training (76K steps, 14.9 hours), the best checkpoint (step 74K) achieves **held-out PPL of 7.13, 25.3% CSR** (95% CI [24.0%, 26.5%]), and **76% MorphAcc**. A key methodological finding emerges: **PPL is the only metric that consistently produces coherent, reliable signal** for checkpoint ranking (7.56 -> 7.17 -> 7.13, all 14 evaluation files agree). The autocompleteness-specific metrics proved fragile: CSR requires token-ID prompts to avoid BOS/tokenizer divergence bugs (string prompts gave ~4% vs ~28%), produces overlapping confidence intervals at n=300, and the blinded human A/B (p=0.82) is underpowered at n=30. A sentence-level typing simulation reveals a **CSR paradox**: the model’s native Basque achieves the lowest simulated CSR (19.2%), below English (26.3%) and Spanish (30.9%) which represent <1% of training data — a structural artifact of agglutinative word length, not a model deficiency. A **CSR inversion** is also documented: next-word CSR does not track PPL improvement across the training trajectory.

We document five **inference engineering strategies** — retokenization fallback, sticky merge, top-k exceeding display-k, next-word candidate extraction, and completion logging with replay — that address failure modes unique to deploying subword-tokenized models as interactive keyboards in agglutinative languages. A cross-model comparison using **Bits Per Character (BPC)** shows Morpheus (91M, BPC 0.970) matching GPT-2 eus-euscrawl (124M, BPC 0.981) and approaching Latxa-Qwen3.5-2B (1.88B, BPC 0.822) at 1/20th the parameters. A data scaling analysis finds the model converged at ~8.8B tokens due to **data quality constraints** rather than quantity, suggesting that for low-resource languages, aggressive quality filtering of a 2–5B token corpus would be more efficient than maximizing raw token count.

1. Introduction

Predictive autocompletion—suggesting the next words as a user types—is a mature technology for high-resource languages. Google’s Gmail Smart Compose serves billions of suggestions daily from an ~80M-parameter LSTM on Cloud TPUs. Gboard provides on-device next-word prediction on smartphones with a 1.4M-parameter, 1.4 MB model. GitHub Copilot applies the same paradigm to code, entirely server-side. However, these systems target languages with simple morphology (English, Spanish, Chinese) where next-word prediction is primarily a collocation problem — and none runs a Smart Compose-equivalent multi-token continuation system entirely on-device.

The research question. Can a Smart Compose-equivalent system run **entirely locally on a consumer laptop**, without any network calls, for use in text editors? This is the primary question Morpheus investigates. A secondary question — whether such a system can also power **mobile next-word prediction** (the Gboard paradigm) — requires a much smaller model; our 91M model serves as a starting point for that trajectory rather than a finished mobile solution. See Section 2.1 for a detailed comparison of production systems.

Basque (Euskara) is fundamentally different. As a language isolate with agglutinative morphology, a single Basque verb can encode subject, direct object, indirect object, tense, mood, and aspect through suffix chains (e.g., *ikusiko zenizkidakeen* — “you would have been able to see them to me”). A Basque noun takes 12+ case suffixes, plus number and definiteness marking (e.g.,

etxeetaraino — “up to the houses”). This means that predicting the next word is not just about collocations—it requires **morphological productivity**: the ability to generate grammatically correct suffix sequences that the model has never seen as a unit during training.

Recent work on agglutinative language modeling confirms this challenge. QuechuaTok (Contreras, 2026) showed that standard BPE tokenizers achieve only 6.67% morpheme boundary accuracy on Quechua, while morphology-aware tokenization reaches 83.33%. Lane et al. (2022) demonstrated that morph-based word completion for Plains Cree requires explicit morphological segmentation to be usable.

Morpheus is designed for this challenge. We build an on-device predictive autocompletion system for Basque that supports two complementary prediction paradigms — **multi-token continuation** (Smart Compose-style inline ghost text for desktop use, accepted with Tab) and **next-word prediction** (smartphone keyboard-style discrete word chips, accepted with a tap) — and that:

1. Runs **entirely locally** on consumer hardware (no cloud dependency, privacy-first)
2. Provides suggestions within $\leq$ **50ms P90 latency** on a standard x86-64 CPU
3. Achieves **meaningful keystroke savings** for practical utility
4. Handles **Basque morphology** — not just memorized collocations but productive case suffix prediction
5. Supports **Euskañol** (Basque-Spanish code-switching), common in informal communication
6. Fits within $\leq$ **300 MB** on disk (feasible for browser extensions and desktop applications)
7. Supports **both prediction paradigms** with inference engineering tailored to each

This paper documents our architecture selection process, training pipeline, evaluation methodology, and final results after complete training (76K steps, best checkpoint step 74K).

2. Related Work

2.1 Predictive Autocompletion

Two prediction paradigms dominate real-world autocomplete systems. **Multi-token continuation** (Smart Compose, Copilot) generates inline ghost text accepted with Tab. **Next-word prediction** (Gboard) offers discrete word chips accepted with a tap. They differ not only in UX but also in deployment constraints, model scale, and engineering challenges. Table 1 situates Morpheus among production systems.

Table 1: Production autocomplete systems

System	Params	Size	Training Data	Deployment	Architecture	Paradigm
Gboard (on-device NWP)	1.4M	1.4 MB	Federated (per-user)	On-device (mobile)	LSTM	Next-word
Gmail Smart Compose	~80M	Server-side	~8B emails (~320B+ tokens est.)	Cloud TPU (data center)	LSTM	Multi-token
GitHub Copilot	Multi-billion	Server-side	Proprietary code corpus	Cloud GPU (Azure)	Transformer (FIM)	Multi-token

System	Params	Size	Training Data	Deployment	Architecture	Paradigm
GPT-2	124M	~50 MB	~423M tokens	On-device (desktop)	Transformer	Multi-token
eus-euscrawl		(Q4)				
Morpheus 91M v2		55 MB	Latxa Corpus v2 (curated, 10B tok)	On-device (laptop)	Mamba-2	Both
		(Q4_K_M)				
Latxa-Qwen3.5-2B	1,882M	~1.2 GB	Latxa Corpus v2 (public, ~4.2B tokens)	High-end only	Qwen3.5 (instruct)	Multi-token

The KV-cache insight. Despite Transformers achieving better perplexity, Google chose an LSTM for Smart Compose because Transformer self-attention requires maintaining keys and values from all previous decoding steps, making per-step latency grow with context length (Chen et al., 2019). GitHub Copilot, also Transformer-based, requires an elaborate global proxy infrastructure (HTTP/2, request cancellation, streaming, geographic routing) to achieve <200ms latency (Cheney, 2025). Google solved the latency problem with data-center TPUs; Morpheus solves it with Mamba-2’s $O(1)$ per-step inference at the architecture level — enabling the Smart Compose paradigm to run **on-device** on a consumer laptop (91M, 55 MB, zero network calls), comparable in scale to Smart Compose’s 80M but without the data-center dependency.

Data asymmetry. Smart Compose’s training data has two advantages unavailable to Basque: **volume** (~8B emails, ~30–60x our 10B token corpus) and **domain match** (trained on emails, deployed for writing emails — the distribution the model learns is exactly what users produce). Morpheus trains on general Basque prose (Wikipedia, news, literature) but deploys as a general-purpose text editor, producing corpus-induced artifacts (Section 6.11) where the model over-predicts encyclopedic patterns. For Basque, no large email or conversational corpus exists — data quality is the binding constraint, not quantity (Section 6.8).

Mobile vs. desktop. The 91M model is sized for the desktop multi-token continuation use case (comparable to Smart Compose). The mobile next-word prediction paradigm (Gboard: 1.4M/1.4MB) requires a distilled variant (~5–10M) that has not yet been trained; the inference engineering strategies in Section 5.5 are designed to carry over.

Trnka & McCoy (2008) defined **Keystroke Savings Rate (KSR)** as the gold standard for word prediction evaluation: the percentage of keystrokes saved by accepting predictions. We adapt this as **Character Savings Rate (CSR)** — a simulation-based metric following the free-acceptance model where accepting a suggestion costs one keystroke (Tab).

2.2 Agglutinative Language Modeling

QuechuaTok (Contreras, 2026) introduced MorphAcc — morpheme boundary accuracy — showing that standard tokenizers radically underperform on agglutinative languages. BPE achieves fertility 1.636 but only 6.67% MorphAcc on Quechua. We adopt MorphAcc for Basque evaluation and replicate the vocabulary-size finding: our 4K Unigram tokenizer achieves 66.7% MorphAcc consistency (vs 28.6% at 32K), mirroring QuechuaTok’s 4K result of 66.67%.

Plains Cree word completion (Lane et al., 2022) used a finite-state morphological analyzer (FST) to segment Cree words into morphemes before prediction. They found that morphological

segmentation is both the input representation and the evaluation target for agglutinative languages. Their KSR improvements (15-30% over non-morphological baselines) motivate future work on morphology-aware tokenization for Morpheus.

Euskarazko LLM-ak (Basque LLMs): The HiTZ center has released Latxa (Llama-2/3-based, 7B-70B) and Orai NLP has released Kimu (Gemma-2-based, 2B-9B). EvalEU benchmark (itzune.eus/evaleu, 2026) shows Kimu 9B outperforming Latxa 8B on text-relevant tasks (XNLI 74.2% vs 56.7%, EusProficiency 51.2% vs 46.3%). While these models demonstrate strong Basque language modeling, their sizes (8B-70B params) are incompatible with on-device deployment. Latxa and Kimu also inherit their parent models’ tokenizers (32K and 256K respectively) without published Basque-specific tokenizer ablations — a gap our vocabulary-size experiment addresses. For baseline comparison (Section 6.7), we evaluate two HiTZ models at scales comparable to or larger than Morpheus: **HiTZ/gpt2-eus-euscrawl** (GPT-2 small, 124M, trained on the EusCrawl corpus) and **HiTZ/Latxa-Qwen3.5-2B** (1.88B, a Qwen3.5-based instruct model adapted for Basque using the publicly available Latxa Corpus v2, ~4.2B tokens; Sainz et al., 2025). The former represents the small-model regime; the latter represents what a 20x larger instruct-tuned model achieves on the same evaluation corpus.

2.3 State Space Models

Mamba (Gu & Dao, 2023) introduced Selective State Space Models as an alternative to Transformers, offering linear-time inference with constant memory — no KV cache. Mamba-2 (Dao & Gu, 2024) reformulated the architecture as Structured State Space Duality, achieving 6-8x Transformer training speed with multi-head SSM support.

LFM-2.5-230M (Liquid AI, 2026) demonstrated SSM viability on edge devices: a 230M hybrid Mamba-MLP model running at 42 tok/s on a Raspberry Pi 5 and 213 tok/s on a Galaxy S25 Ultra, within 400 MB memory. This validates the on-device deployment path we pursue.

2.4 Evaluation Methodology for Autocomplete

Chai-TeA (2024) is the closest analog to our evaluation: a large-scale benchmark for Chinese input methods with 26K+ test prefixes and a **saved@k** metric. They attempted LLM-as-judge for semantic matching of alternative completions but found it “very challenging” and deferred it — confirming the multiple-valid-completions problem we encounter.

Kosyak & Tyers (2022) evaluated predictive text for agglutinative languages using FST-based models and KSR with user studies, confirming that the multiple-valid-forms problem is central to agglutinative autocomplete evaluation.

WSTypist (2026) used simulation-based mobile typing evaluation, confirming that simulation metrics are accepted practice when user studies are infeasible, provided they are at scale with a realistic acceptance model.

3. Architecture Selection

3.1 Problem Constraints

Our constraints create a tight optimization problem:

Constraint	Target	Why
Inference latency	P90 $\leq$ 50ms per token	Users perceive $>$ 100ms as lag
Memory footprint	$\leq$ 300 MB on disk, $\leq$ 500 MB RAM	Must fit consumer devices + browser extensions
No network calls	Zero external dependencies	Privacy requirement; all data stays local
Basque morphology	Correct case suffix prediction	Core user need, not just collocations
Training budget	1x NVIDIA L40 (48 GB), 5-10 days	Realistic for a small research team
Code-switching	Basque-Spanish (Euskarañol)	$\sim$ 30% of informal Basque text contains Spanish

3.2 Candidate Architectures

We evaluated three candidate architectures against these constraints:

Path A: xLSTM (Modern Recurrent)

- 50-100M parameters
- $O(1)$ per-step inference, constant memory, no KV cache
- Training from scratch on curated Latxa Corpus v2 (Section 4.2)
- ONNX Runtime INT8 deployment
- **Pros:** Safest latency, reuses v1 infrastructure, predictable
- **Cons:** No pre-trained Basque knowledge, lower expected quality ceiling

Path B: Distilled Transformer (from Kimu 9B)

- 200-500M parameters (pruned + distilled from Orai Kimu 9B)
- $O(n)$ inference with KV cache
- Best expected quality due to pre-trained Basque knowledge
- llama.cpp GGUF deployment
- **Pros:** Highest potential quality (EvalEU-validated teacher), mature ecosystem
- **Cons:** KV cache adds latency unpredictability, tokenizer mismatch (Gemma 256K $\rightarrow$ must prune), most engineering complexity

Path C: Mamba-2 SSM

- 100-300M parameters
- $O(1)$ per-step inference, constant memory, no KV cache
- Parallelizable training (scan operation)
- llama.cpp GGUF deployment
- **Pros:** Best quality-latency balance, future-proof (SSMs are the on-device trajectory), LFM-2.5 validates edge feasibility

- **Cons:** Newer, less battle-tested ecosystem; ONNX support immature (use GGUF instead)

3.3 Decision: Mamba-2 (Path C)

We selected Mamba-2 for the following weighted scoring:

Criterion	Weight	xLSTM	Distilled Transformer	Mamba-2
Expected prediction quality	30%	3	5	4
Inference latency safety	25%	5	3	5
Engineering effort / risk	20%	4	2	3
Ecosystem maturity	10%	3	5	3
Future-proofing	10%	3	4	5
Training cost	5%	5	3	4
Weighted Score		3.80	3.70	4.00

Mamba-2 combines the LSTM-like inference properties essential for on-device autocomplete (constant memory, predictable latency) with dramatically better language modeling capacity than recurrent architectures at the same scale. This directly mirrors Google’s Smart Compose decision (Section 2.1): both chose recurrent over Transformer architectures to avoid KV-cache latency, but Mamba-2 solves it at the architecture level rather than with data-center TPUs.

The decision to not pursue Path B (distillation) was driven by practical concerns: the Gemma tokenizer’s 256K vocabulary would require aggressive pruning for a 200M model (embedding table alone = 524 MB), and KV cache management on consumer CPU introduces latency variance that violates our 50ms P90 constraint.

4. Model Architecture & Training

4.1 Model Configuration

Parameter	Morpheus-Small
Architecture	Mamba-2 (pure SSM)
d_model	768
n_layer	24
d_state	64
expand	2
headdim	64
Total params	~91M
Vocabulary size	4,000
Seq length (training)	1024
Batch size	64 (x2 accumulation)
Effective tokens/step	131,072
Learning rate	2e-3 (cosine decay to 1e-5)
Warmup tokens	50M
Total training tokens	10B

Parameter	Morpheus-Small
Total steps	76,294

At 4K vocabulary, only 3.4% of parameters are in the embedding table (3.07M of ~91M), compared to 27% at 32K. This parameter efficiency is a key advantage of the small-vocabulary approach for compact models.

4.2 Data Pipeline

Corpus: 4.62 billion subword tokens (pre-tokenized as uint16 .npy, 9.24 GB) from 11 curated Basque text sources. The corpus is derived from the publicly available **Latxa Corpus v2** (HiTZ/latxa-corpus-v2 on HuggingFace; Etxaniz et al., 2024), with HPLT v1 excluded (83.8% duplicates) and additional deep-cleaning (Phase 0.5 + Phase 3: normalization, deduplication, quality filtering). The 11 retained sources are EusCrawl v2 (web crawl, cleanest), HPLT v2 (web crawl, replaced v1), CulturaX, FineWeb2, FinePDFs, Colossal OSCAR, ZelaiHandi (news/blog), ParlEus (parliament transcriptions), Wikipedia Basque (used for validation split), and two official gazette sources (BOPV, BOTHA). The gazette sources are included for training but excluded from tokenizer training due to >60% duplicate rates that would bias the vocabulary toward gazette boilerplate. Training saw ~10B tokens total (~2.16 epochs over the 4.62B-token corpus).

Cleaning: Four-phase data cleaning pipeline applied before tokenization: (1) document re-parsing (encoding normalization, corrupted document detection), (2) form regularity (punctuation boundaries, whitespace normalization), (3) content filtering (validation/test leakage removal, line length filters, outlier removal), (4) deduplication (exact + near-deduplication via MinHash LSH).

Corpus-quality audit (2026-07-03): A full-source audit revealed quality disparities. Four sources were excluded: **hplt-v1** (83.8% duplicate rate, 4.9% Basque signal — replaced by **hplt-v2**), **B0G** (Phase 2 sentence splitting destroyed legal text into mid-sentence fragments), **Aldizkariak** (35% boilerplate), and **BERnaT BSM** (dialectal, non-standard orthography). The remaining 11 sources (128M lines, average quality 4.6/5 from LLM-based audit) proceed to training.

Validation leakage prevention: 68,755 lines from the held-out validation set (`wiki_valid.txt`) are excluded from training pretokenization via `--exclude-lines-file`, ensuring zero overlap between training and evaluation data.

Tokenizer: SentencePiece Unigram with **4,000-token vocabulary**, trained on 9 sources (all 11 except BOPV and BOTHA). See Section 4.5 for the vocabulary-size ablation that motivated this choice.

4K tokenizer quality (verified 2026-07-04): fertility 2.52 tokens/word, 100% roundtrip fidelity, 99.95% character coverage, 107-character alphabet. All core agglutinative patterns split cleanly: `etxe+a -> |etxe a` (house + the), `etxe+tik -> |etxe tik` (house + from). Multi-layer morphology also resolves: `mendikoak -> |mendi ko ak` (mountain + of + the-plural), three distinct morphemes. Compare this to the 32K tokenizer which fused entire inflectional clusters into opaque tokens like `|etxetik`, `|etxera` — see Section 4.5 for the full comparison.

4.3 Training

Training was performed on a single NVIDIA L40 GPU (48 GB GDDR6, Ada Lovelace) with the following setup:

- **Framework:** PyTorch 2.4+ with `mamba-ssm` and `causal-conv1d` CUDA kernels
- **Precision:** bfloat16 (BF16) for training stability
- **Optimizer:** AdamW (beta_1=0.9, beta_2=0.95, weight_decay=0.1)
- **Schedule:** Linear warmup (50M tokens) -> cosine decay to 1e-5
- **GPU config:** Power limit 260W, persistence mode enabled
- **Throughput:** ~55k tok/s (settled)
- **W&B run:** knid3x95 (“rich-forest-14”), project `morpheus-v2-mamba`

Training progress (4K model, W&B run knid3x95):

Milestone	Step	Valid Loss	Valid PPL	Tokens Seen	% Complete
Resume point	16,000	—	—	2.10B	21.0%
Checkpoint	32,000	2.0229	7.56	4.19B	41.9%
Mid-training	45,000	1.9864	7.28	5.90B	59.0%
Evaluation	54,000	1.9698	7.17	7.08B	70.8%
Best (converged)	74,000	1.9641	7.13	9.70B	97.0%
Final	76,294	1.9641	7.13	10.0B	100%

Validation loss decreased monotonically throughout training (2.05 -> 1.96), with PPL improving from 7.8 (step 25K) to 7.13 (step 74K). The model is fully converged — validation loss is flat from step 67K onward (Delta < 0.001). The best checkpoint (step 74K, valid_loss=1.9641, PPL=7.13) is used for all final evaluations and deployed models. Step 76K (final step) achieved identical PPL (7.13) and is not distinguishable in quality. The training budget of ~10B tokens yields a tokens-to-parameters ratio of approximately 110:1 — below the Mosaic inference-optimal (190:1) and MiniCPM small-model-optimal (192:1) recommendations, indicating the data budget is reasonable by modern small-model standards; see Section 6.8 for a detailed scaling-law analysis.

Checkpoint integrity: Checkpoints are saved atomically (write to `.tmp` then `os.replace()`) every 2,000 steps. A file-based stop monitor detects checkpoint completion via size-stability polling (3 consecutive unchanged polls + size >= 540MB) and sends SIGINT for clean W&B flush.

4.4 Pre-Training Validation: The Three-Gate Protocol

Perplexity is the standard language modeling metric, but it is insufficient as a sole quality gate for autocomplete: a model can achieve low PPL while producing degenerate suggestions if the training corpus contains systematic artifacts — for example, bare numbers in sentence position (decree IDs, budget amounts from official gazette texts) that bias the model toward numeric predictions instead of Basque words. This limitation is explored in detail in Section 6, where PPL improves while the multi-token autocomplete metrics lack the statistical power to confirm the improvement at current sample sizes.

To ensure data and pipeline integrity before committing GPU resources, we designed and executed a three-gate pre-training validation protocol. **No training run may proceed to the L40 without passing all three gates.**

Gate 1: Corpus Content Audit (CPU, ~30 min). An LLM-based quality audit using DeepSeek-V4-Pro evaluates 40 random lines per source, classifying each by text type, flagging quality issues, and rating Basque quality on a 1–5 scale. The 2026-07-04 audit flagged BOG (2% clean) and

aldizkariak (38% clean). Both were excluded. The remaining 11 sources scored an average quality of 4.6/5.

Gate 2: Proxy Overfit Test (GPU, ~20s). A canary test: a 0.7M-parameter Mamba-2 model (128x smaller than target) attempts to memorize 5 hand-crafted Basque sentences not in the training corpus. If it can memorize novel Basque from the tokenized .npy format in 300 steps, the pipeline (tokenizer, serialization, architecture, training loop) is proven sound. Any downstream failure must then come from data quality or training scale, not infrastructure.

Metric	Value
Initial loss (step 0)	8.30 (random initialization)
Final loss (step 300)	0.049 (170x reduction)
Canary token accuracy	57.7% ($\geq 50\%$ threshold)
Runtime	18.5 seconds
NaN events	0

Gate 3: Autocomplete Smoke Test (GPU, ~5 min). Evaluates whether the full 91M model produces useful autocomplete suggestions on real Basque text. **Passed:** at step 32K, the model achieves CSR=24.9% on 300 held-out sentences and MorphAcc=70%, confirming the model produces coherent Basque completions.

Gate	Runtime	What it validates	clean-v3 result
1: LLM audit	~30 min CPU	Source quality, fragments, boilerplate	11/13 sources pass
2: Proxy overfit	~20s GPU	Tokenizer integrity, data format	[OK] 57.7% canary accuracy
3: Autocomplete smoke	~5 min GPU	Autocomplete quality on real text	[OK] CSR=24.9%, MorphAcc=70%

Each gate targets a distinct failure class that PPL alone cannot detect: data contamination and source quality (Gate 1), pipeline and serialization integrity (Gate 2), and inference-time autocomplete behavior (Gate 3). Together they ensure that low PPL reflects genuine language modeling quality rather than artifacts of data or infrastructure.

4.5 Tokenizer Strategy: Deep Research and Implications

The choice of tokenizer is the single most consequential design decision for agglutinative language modeling. Our tokenizer research spanned seven recent papers (2024–2026) covering 70+ languages and multiple agglutinative language families.

4.5.1 Six Papers That Reshaped Our Understanding Our tokenizer research spanned six recent papers (2024–2026) covering 70+ languages and multiple agglutinative language families. Three findings converge:

1. **Fertility is misleading for agglutinative tokenizers.** QuechuaTok (Contreras, 2026) — the most directly relevant work — evaluated BPE, Unigram, WordPiece, and morphology-aware PRPE on Southern Quechua (a suffixing agglutinative language structurally analogous to Basque). BPE achieves the lowest fertility (1.636) by memorizing entire polymorphemic words as single tokens, yet achieves only **6.67% MorphAcc**. Unigram at 4K achieves **66.67% MorphAcc**, dropping to 26.67% at 8K. We replicate this finding for Basque (Section 4.5.4) and extend it: **fertility is not just insufficient — it is a *confound* that rewards the very behavior that destroys morphological generalization.**
2. **Unigram > BPE for agglutinative languages.** Xu & Kim (2026) found Unigram consistently outperforms BPE on POS tagging across six Uralic languages. Stephen & Libovický (2026) confirmed Unigram > BPE > WordPiece for morphological alignment, with smaller vocabularies yielding better alignment.
3. **Morphological pre-segmentation improves downstream performance.** García et al. (2025) showed that training BPE on morphologically pre-segmented text improved masked LM performance for Spanish. Hu (2025) found word-level tokenization outperformed BPE for morphologically rich languages under low-resource conditions.

A notable counterpoint: Arnett et al. (2025) found that morphological alignment explains only a small fraction of variance ($R^2 = 0.005\text{--}0.024$) in downstream performance across 70 languages — for Basque specifically, MorphScore precision was 0.11–0.12, among the lowest of all languages evaluated. This suggests that while morphological alignment is necessary, it is not sufficient for downstream quality at scale.

4.5.2 What Basque LLMs Chose Latxa (HiTZ, 2024): The dominant Basque LLM family (7B–70B) uses Llama 2’s BPE tokenizer (32K vocabulary) without modification. The Latxa paper does not discuss tokenizer choices.

Kimu (Orai NLP, 2025): The Gemma-2-based Basque model family inherits Gemma’s SentencePiece tokenizer (256K vocabulary).

The open question: Neither Latxa nor Kimu has published an ablation study on tokenizer impact for Basque. At 7B–70B parameters, models overcome tokenizer deficiencies through scale — but our 91M model cannot afford a suboptimal tokenizer.

4.5.3 Retrospective on Our Decision Our v1 tokenizer decision chose SentencePiece Unigram at 32K based on fertility 1.71. The 2026 research reveals:

1. **Unigram was correct over BPE.** Xu & Kim (2026) and Stephen & Libovický (2026) support Unigram for agglutinative settings.
2. **32K vocabulary places us in the surface-form memorization regime.** At 32K, the tokenizer memorizes frequent wordforms as atomic units, fragmenting morphemes arbitrarily.
3. **No morphological pre-segmentation was the critical omission.** The literature converges: high morpheme-boundary accuracy requires injecting morphology into tokenizer training.

4.5.3.1 The Morfessor Attempt and Pivot An initial Morfessor 2.0 pre-segmentation attempt failed due to poor segmentation quality on mixed-language text (the MDL objective cannot distinguish Basque morphology from arbitrary character sequences in foreign words). We pivoted to testing vocabulary size as the primary variable — a simpler experiment with a stronger literature basis. Details in Appendix A.

4.5.4 The Vocabulary Ablation Experiment

We formulated a testable hypothesis:

The morphological boundary accuracy of a SentencePiece Unigram tokenizer degrades monotonically with vocabulary size because larger vocabularies accumulate frequent surface forms as atomic tokens, fragmenting morpheme boundaries arbitrarily.

Metric: MorphAcc consistency. For each test word with a known root–suffix boundary (e.g., `etxe|tik`), we check whether the tokenizer places a token boundary at the morpheme boundary. A word scores `boundary_correct = true` if and only if the root and suffix are in **separate tokens** — not merely substrings of a single fused token. For example, `etxetik` -> `|etxe tik` scores [OK] (boundary preserved); `etxetik` -> `|etxetik` scores [X] (fused into one token). The aggregate metric is the percentage of test words with `boundary_correct = true`.

Experimental design. We trained SentencePiece Unigram tokenizers at 4K, 8K, 16K, and 32K on a proportional sample from the full corpus (336.8 MB, 3.5M lines, ~84M tokens, drawn proportionally from all 15 source files). All tokenizers used identical SentencePiece parameters: `model_type=unigram`, `character_coverage=0.9995`, `byte_fallback=True`. Training took ~60s per tokenizer on CPU. The 32K baseline was our existing production tokenizer trained on the same corpus with the same parameters.

Test words. We evaluated 21 Basque words covering five roots (`etxe`, `lagun`, `mendi`, `gizon`, `kale`) and eight case suffixes (absolutive `-a`, allative `-ra`, ablative `-tik`, genitive locative `-ko`, inessive `-an`, comitative `-arekin`, benefactive `-arentzat`, causal `-arengatik`). Representative examples:

Root	Suffix	Test word	Expected boundary
etxe	-a	etxea	etxe\ a
etxe	-tik	etxetik	etxe\ tik
etxe	-arekin	etxearekin	etxe\ arekin
mendi	-ra	mendira	mendi\ ra

(The full 21-word list with per-tokenizer results is available in the supplementary materials.)

Results:

Tokenizer	Vocab Size	Fertility	MorphAcc Consistency	Change
baseline-32k	32,000	1.85	28.6% (6/21)	baseline
raw-16k	16,000	2.06	52.4% (11/21)	+23.8pp
raw-8k	8,000	2.28	61.9% (13/21)	+33.3pp
raw-4k	4,000	2.58	66.7% (14/21)	+38.1pp

The hypothesis was confirmed with striking precision:

1. **The QuechuaTok finding replicates for Basque.** The 4K MorphAcc (66.7%) is nearly identical to QuechuaTok’s 4K Unigram result (66.67%).
2. **The degradation is monotonic and accelerating.** Each doubling of vocabulary costs progressively more: 4K->8K loses 4.8pp, 8K->16K loses 9.5pp, 16K->32K loses 23.8pp.
3. **Fertility is the mechanism, not the metric.** The 4K tokenizer has the worst fertility (2.58 vs 1.85 for 32K) because it is forced to decompose — and that forced decomposition is precisely what preserves morpheme boundaries.

4. **4K is the sweet spot for Basque at 91M parameters.** At 4K, every root is a separate token from every case suffix, and verbal agreement morphemes are independently accessible.

4.5.4.1 Beyond Nominal Morphology: The Verbmorph Gap The MorphAcc consistency metric in Section 4.5.4 tests **nominal morphology** — root + case suffix. The decisive difference between 4K and 32K emerges even more starkly in **verbal morphology**, where Basque’s polysynthetic verb structure encodes subject, object, indirect object, tense, and mood in a single word:

Verb form	Gloss	4K tokenization	32K tokenization
dizkizut	<i>I have them to you</i>	di zki zu t	dizkizut
dakizkioke	<i>he can know them to him</i>	da ki zki o ke	dakizki ok e
zitzaizkidan	<i>they were to me</i>	zitzaizki dan	zitzaizkidan

At 4K, the pluralizer **zki** is an independent reusable token present in all three verbs. A Mamba-2 model can productively recombine **di**, **zki**, **zu**, **t** to form unseen verb inflections. At 32K, **zki** is buried inside opaque atomic tokens that share no subword structure.

This is the **fertility paradox**: lower fertility (fewer tokens per word) is achieved exactly by fusing morphemes into opaque units. The fertility metric favors the very behavior that destroys morphological generalization. **Low fertility is the *mechanism* of the surface-form memorization regime, not a desirable property.**

4.5.4.2 Where 4K Still Fails The 4K tokenizer is not perfect: it struggles with (1) **multi-layer suffixes** where a case suffix itself decomposes into multiple morphemes (e.g., **etxearentzat** -> |etxe a rentzat instead of **etxe|arentzat**), and (2) **epenthetic vowels** where Basque inserts **e-** before consonant-initial suffixes (e.g., **lagunetik** -> |lagun etik instead of **lagun|tik**). These remaining failures motivate the Apertium pre-segmentation future work (Section 7.2). Full failure-mode table in Appendix B.

4.5.5 Downstream Perplexity Confirmation The vocabulary ablation was further validated by downstream perplexity evidence from the QuechuaTok study: 4K Unigram achieves the lowest downstream perplexity among vocabulary sizes tested, confirming that the morphological alignment advantage translates to better language modeling, not just better MorphAcc.

At our model scale, the parameter-efficiency argument is also decisive: at 4K vocab, only 3.4% of the 91M parameters are embeddings (vs 27% at 32K), freeing capacity for the SSM layers that drive sequence modeling quality.

5. Deployment Pipeline

5.1 Export: PyTorch -> HuggingFace -> GGUF

Direct ONNX export of Mamba models is not viable for production CPU inference. Instead, we use `llama.cpp`'s native Mamba-2 support:

1. PyTorch checkpoint -> HuggingFace format (custom export script)
2. HuggingFace -> GGUF FP16 (`llama.cpp convert_hf_to_gguf.py`)
3. GGUF FP16 -> Q4_K_M quantization (`llama-quantize`)

The export script writes `tokenizer_config.json` with `add_bos_token: false` and a `generation_config.json`, ensuring `llama.cpp` does not auto-prepend a BOS token. This fix was motivated by a debugging experiment: the model was trained without BOS, but `llama-server` auto-prepended BOS for string prompts, producing CSR of ~4% (vs ~28% with correct token-ID prompts). This was diagnosed as two compounding issues — (1) BOS mismatch (the model never saw BOS during training) and (2) `llama.cpp`'s built-in `SentencePiece` tokenizer diverging from the reference `sentencepiece` library on this 4K vocabulary, causing incorrect tokenization of long words. The fix is two-layer: the export writes `add_bos_token=false`, and the demo sends token IDs (not strings) to `llama-server` (Section 5.4).

5.2 Quantization

Format	Size	BPW	Notes
FP32 (PyTorch checkpoint)	~547 MB	32	Training only
FP16 (GGUF)	181 MB	16	Reference quality
Q8_0 (GGUF)	97 MB	8	Near-lossless
Q5_K_M (GGUF)	64 MB	5.5	High quality
Q4_K_M (GGUF)	55 MB	4.92	Deployment default

5.3 Inference

On the NVIDIA L40 (GPU), the f16 model generates at ~550 tok/s in batch. On consumer x86-64 CPU (Intel Xeon, 8 threads), the Q4_K_M quantized model achieves ~250 tok/s in batch inference via `llama-server`. Interactive single-token generation is estimated at ~40-80 tok/s.

5.4 Demo Server

A Docker-based demo server provides: - **Greedy mode** (Smart Compose style): `temperature=0.0`, deterministic suggestions - **Sampling mode**: configurable temperature, `top_p`, WebSocket streaming - **Token-ID prompts**: Sends token IDs (not strings) to `llama-server`, bypassing the BOS/tokenizer divergence documented in Section 5.1 - **Smart context**: strips trailing subword fragments so the model sees only complete tokens - **Ghost suffix**: deduplication of user-typed text from model prediction for inline ghost text display - **Output filtering**: punctuation collapse, U+FFFD removal (for undertrained model artifacts) - **Model hot-reload**: `POST /api/model/reload` swaps GGUF files without container restart

5.4.1 Two Prediction Paradigms

The demo implements two distinct prediction paradigms, which we distinguish throughout the evaluation because they have different user experiences, failure modes, and applicable metrics:

	Multi-token continuation	Next-word prediction
Metaphor	Smart Compose (Gmail)	Predictive keyboard (smartphone)
Morpheus model	91M (55 MB), on-device	91M — needs distillation for mobile
Output	N tokens of gray inline text	3 discrete word chips
Acceptance	Tab accepts entire continuation	Tap selects one word
Failure mode	Repetition loops, ghost-text jitter	Tokenization trap, prediction vanishing
Evaluated by	CSR, Human A/B (Section 6)	Completion logging + replay (Section 5.5.6)

Multi-token continuation is the simpler inference case: the model greedily extends the context. Next-word prediction is harder because it must produce *whole words* as discrete options, exposing the tokenization trap (Section 5.5.1). This distinction maps to evaluation: CSR and human A/B measure multi-token quality, while completion logging measures next-word quality. Conflating them would obscure why PPL improvements are not confirmed by CSR while the keyboard experience benefits from candidate carry-forward.

5.5 Inference Engineering for Agglutinative Keyboards

This section addresses the **next-word prediction** paradigm (Section 5.4.1): deploying a language model as a real-time predictive keyboard that offers whole-word suggestion chips. This is the harder of the two paradigms because it must produce discrete, complete words — which exposes the tokenization trap, a structural failure mode of subword tokenization that does not appear in multi-token ghost-text continuation or in batch evaluation. Each strategy below is motivated by a concrete failure observed during development. The strategies are architecture-agnostic and would carry over directly to a future distilled mobile model (Section 2.1).

5.5.1 The Tokenization Trap In an agglutinative language with a 4K subword vocabulary, the same word can be reachable through multiple token paths — and the path the user’s partial input lands on may not reach the correct completion.

Example: The Basque word *Kaixo* (“hello”) tokenizes as [*|Ka*, *i*, *xo*]. But when the user types *Kaix*, the tokenizer segments it as [*|Ka*, *ix*] — a different path that cannot reach the *xo* token. The model may know the word perfectly, but the greedy continuation from [*|Ka*, *ix*] produces *Kaixan*, *Kaix-*, *Kaixko* — never *Kaixo*.

This is not a model deficiency; it is a structural artifact of subword tokenization. The problem is especially acute in agglutinative languages because long, morphologically complex words have many possible segmentation paths, and short prefixes often land on the wrong one.

5.5.2 Retokenization Fallback Strategy: When generating word-completion candidates, query the model from progressively shorter prefixes in parallel, then filter results by the user’s actual typed prefix.

For input *Kaix*, the system queries three paths simultaneously:

Path	Prefix	Token IDs	Can reach <i>Kaixo</i> ?
0	Kaix	[Ka, ix]	[X] (wrong path)
1	Kai	[Ka, i]	[OK] (xo is reachable)
2	Ka	[Ka]	[OK] (but noisier)

Path 1 reaches [|Ka, i], where the model predicts *xo* at 54.2% probability. The result *Kaixo* passes the `startswith("Kaix")` filter and surfaces as a candidate. All paths fire in parallel via `asyncio.gather`, keeping latency at ~1x a single call rather than 3x.

A **from-scratch path** (empty prefix, predicting the next word from preceding context only) rescues single-token whole words. For example, *bezala* is a single token (|bezala); when the user types *b*, the token |b cannot reach |bezala. Querying from the preceding context (“*Ni ondo, beti*”) surfaces *bezala* as a next-word prediction, which passes the `startswith("b")` filter.

5.5.3 Sticky Merge: Candidate Carry-Forward Problem: When the model predicts a next word (e.g., *izan* after *idatzia*) and the user types the first letter (*i*), the system switches from next-word prediction to word-completion mode. The token path changes (|*izan* is one token, but |*i* + continuation is a different path), and the previously good prediction vanishes from the candidate list — even though the user is typing exactly the word that was predicted.

Strategy: Maintain a *sticky pool* of the previous render’s candidates. When new candidates arrive, filter the sticky pool by the current typed prefix. Survivors are merged with fresh candidates, receiving a small probability boost (+0.1) to compensate for the fact that cross-path probabilities (next-word vs. word-completion) are not directly comparable.

State 1: "...idatzia" -> [dago, dezakezu, daiteke, behar, izan]
 (izan at rank 5, not visible in top-3 chips, but stored in sticky pool)

State 2: "...idatzia i" -> fresh: [iristeko, itxa, ikur, ...]
 Sticky survivor: izan (prob=0.087, boosted to 0.187)
 Merged result: [iristeko, izan [OK], itxa]

The sticky pool resets on chip acceptance and message send, preventing stale candidates from persisting across word boundaries.

5.5.4 Top-k Exceeds Display-k The keyboard displays 3 suggestion chips but fetches 5 candidates from the server. The extra candidates populate the sticky pool, enabling carry-forward of lower-ranked but relevant predictions. Without this, *izan* (rank 5, prob=0.087) would never enter the sticky pool and could not be rescued in State 2.

5.5.5 Next-Word Candidate Extraction When the model’s greedy continuation at a word-completion level begins with a space (|-prefixed token), it signals that the model considers the current word complete and is predicting the *next* word. Rather than discarding these tokens (as

noise), we extract them as next-word candidates with an `is_next_word` flag. The frontend handles these differently: inserting a leading space before the word, matching the user’s expectation that the current word is finished.

This also handles the edge case where the user has typed a complete word without a trailing space (e.g., “*Kaixo, zer*”): the model predicts *moduz* as a next word, and the candidate appears despite no explicit word boundary.

5.5.6 Completion Logging and Replay Every chip acceptance is logged to a JSONL file with: timestamp, model checkpoint, context, smart context, accepted word and its probability, and all candidates offered. This transforms real user sessions into an evaluation dataset that can be replayed against any checkpoint:

```
python scripts/replay_completions.py --models step_0032000.Q4_K_M step_0054000.Q4_K_M
```

The replay script hot-reloads each checkpoint, queries the same contexts, and checks whether the user-accepted word appears in the top-k. This enabled a critical finding: an apparent model regression (step 54K “forgetting” *Kaixo*) was diagnosed as an **inference engine bug**, not a model deficiency (Section 5.5.7).

The keyboard candidate algorithm (retokenization fallback, sticky merge, top-k fetch, acceptance semantics) is also ported to PyTorch in `src/eval_utils.py` as `evaluate_next_word_csr`, enabling training-time validation that faithfully reflects the deployed demo. This runs natively on the GPU model (no `llama.cpp` dependency) during periodic validation, reporting decomposed metrics (word accuracy, acceptance rate, average prefix length, average confidence) alongside a simulated CSR. It is used as a **secondary metric** — PPL remains primary for checkpoint ranking — and the decomposed metrics avoid the CSR paradox (Section 6.13) because they do not conflate model quality with morphological word length.

The replay system also enabled a critical debugging finding: an apparent model regression (step 54K “forgetting” *Kaixo*) was traced to a stale Docker cache running an older `llama.cpp` with a bug in the SSM scan computation for Mamba-2 (`n_groups > 1`, fixed in commit `dc2187d48`). **When deploying Mamba-2 models with `llama.cpp`, pin to a build that includes this commit (2025-07-04 or later).** Details in Appendix C.

6. Evaluation

6.1 Evaluation Methodology

We employ a multi-metric evaluation suite designed to capture different aspects of autocomplete quality for agglutinative languages. Each metric has known limitations, and we use them in concert to triangulate true model quality.

Metrics

1. **Perplexity (PPL)** — The full next-token distribution quality. Computed on 1.83M held-out tokens (the validation set excluded from training via line-level leakage prevention). This is the smoothest, lowest-variance metric with no exact-match artifact. Computed with training-matching semantics: 1024-token windows, `</s>` separators included in loss, `ignore_index=0` for `<unk>`, no BOS, `bfloat16`.

2. **Character Savings Rate (CSR)** — Simulates keystroke-by-keystroke typing. For each character in the target completion, the model receives `prompt + typed_so_far` and we check if its top-1 greedy prediction aligns with the remaining target. Acceptance costs 1 keystroke (Tab), following Trnka & McCoy (2008). We report **bootstrap 95% confidence intervals** (1000 resamples) on 300 held-out sentences. *This metric measures the **multi-token continuation** paradigm (Section 5.4.1): the model produces a greedy continuation and we check character-level alignment.*
3. **Morpheme Boundary Accuracy (MorphAcc)** — For test cases with known morphological segmentation (e.g., `etxe|tik`), we compute whether the model’s top-5 predictions include a token that respects the morpheme boundary. 50 tests across 5 roots x 4 case suffixes, with and without context.
4. **Case Paradigm Completion** — For each of 6 Basque nouns, test all 14 grammatical cases (84 total). The model receives the bare root and we check if the correct case suffix ranks in top-K.
5. **Blinded Human A/B Evaluation** — 30 fresh prompts from held-out validation text. Both checkpoints generate greedy completions (f16, reference sentencepiece, no quantization confound). A/B randomly assigned per prompt. The Basque expert judges quality (grammaticality, naturalness, autocomplete usefulness) without knowing which checkpoint is which. *This measures the **multi-token continuation** paradigm.*
6. **Completion Logging + Replay** — Real user chip acceptances are logged with full candidate context and replayed across checkpoints (Section 5.5.6). *This measures the **next-word prediction** paradigm.*
7. **Keyboard Simulation (next-word)** — Two variants: (a) a frontend-faithful typing simulation (sticky merge, top-3 chips, acceptance semantics) that types 15 translated sentences (5 Basque, 5 English, 5 Spanish) char-by-char, and (b) a PyTorch-native port of the demo keyboard algorithm (`evaluate_next_word_csr` in `src/eval_utils.py`) that runs during training validation on the same 30 CSR test sentences. Both report decomposed metrics: word accuracy, acceptance rate, average prefix before acceptance, and average confidence — alongside a simulated CSR. *This is a **secondary metric**; PPL remains primary for checkpoint ranking. The decomposed metrics avoid the CSR paradox (Section 6.13) because they do not conflate model quality with morphological word length.*
8. **Bits Per Character (BPC)** — Total NLL in bits divided by character count. **Tokenizer-independent**, enabling fair comparison between models with 4K, 50K, and 248K vocabularies. Used for cross-model comparison (Section 6.7).
9. **Simplified Next-Word CSR (cross-model)** — Greedy decode until a word boundary, extract the first word, compare to target. No inference engineering. Used for fair raw-model comparison across architectures (Section 6.7).

Why multiple metrics No single metric is sufficient for agglutinative autocomplete. In practice, **PPL is the only metric that consistently produces coherent, reliable signal** for checkpoint ranking. The autocomplete-specific metrics proved fragile to implement and underpowered at available sample sizes — we detail why in Section 6.9. The metrics serve as sanity checks rather than ranking tools at this scale.

6.2 Perplexity (PPL)

PPL is computed on two text sets: (1) the held-out validation set (1.83M tokens, genuinely excluded from training), and (2) a real corpus of Wikipedia + Berria articles (140K tokens, **contaminated** — these articles appear in training sources, so absolute PPL is optimistic; the relative comparison remains valid as all checkpoints saw the same text).

Metric	Step 32K	Step 54K	Step 74K	Trend
Held-out valid PPL (clean)	7.56 (loss 2.0229)	7.17 (loss 1.9698)	7.13 (loss 1.9638)	↓ monotonic
Real corpus PPL (contaminated)	10.53 (loss 2.3540)	9.90 (loss 2.2923)	9.83 (loss 2.2853)	↓ monotonic

Per-file consistency: Every single one of the 14 real-corpus files shows monotonic improvement across all three checkpoints, with 32K->54K per-file deltas ranging from -0.54 to -0.89 PPL points and 54K->74K deltas ranging from -0.03 to -0.10. The 54K->74K improvement is small — the model has essentially converged — but it is consistent across all files with no reversals. This is statistically unambiguous.

Note: PPL is not comparable across different vocabulary sizes. The old 32K-vocab model reached PPL=30.78 at step 30K, but this number is not directly comparable to the 4K-vocab model’s PPL=7.56 because the vocabulary size affects PPL (smaller vocab = fewer choices per token = lower PPL). All comparisons here use the *same* 4K vocabulary, so they are valid.

6.3 Character Savings Rate (CSR)

Definition:

$$CSR = 1 - \frac{\text{keystrokes_needed} + \text{acceptance_cost}}{\text{len}(\text{target_completion})}$$

where acceptance_cost = 1 (Tab key to accept the suggestion), following the free-acceptance model of Trnka & McCoy (2008).

Results (300 held-out sentences from `wiki_valid.txt`, seed=20260710, f16 GPU inference):

Metric	Step 32K	Step 54K	Step 74K
Macro CSR	24.90%	25.23%	25.26%
Micro CSR	24.57%	25.03%	25.06%
95% CI (macro)	[23.64%, 26.21%]	[23.98%, 26.48%]	[24.00%, 26.52%]
n_tests	300	300	300

All confidence intervals overlap -> the differences are NOT statistically significant. 74K is directionally the best (agreeing with PPL), but CSR cannot distinguish the three checkpoints

at this quality level. Notably, the 54K->74K improvement in CSR (+0.03pp) is negligible despite continued PPL improvement (7.17->7.13), confirming that CSR saturates once models reach competence.

By target length, short completions (4–6 words) are saturated at all checkpoints (~27%); the 32K->54K improvement concentrates in long targets (10–12 words, +1.39pp) but does not continue to 74K. CSR has saturated.

6.4 Morpheme Boundary Accuracy (MorphAcc)

Results (50 tests, 5 roots x 4 suffixes x 2 conditions [bare + contextual], f16 GPU):

Metric	Step 32K	Step 54K	Step 74K
MorphAcc	70% (35/50)	76% (38/50)	76% (38/50)
Avg boundary prob mass	17.8%	19.4%	19.5%

MorphAcc improved from 32K to 54K (+6pp) but plateaus at 54K — step 74K shows no further improvement (76%, same 38/50 hits). The average boundary probability mass continues to creep up slightly (19.4% -> 19.5%), suggesting marginally more confident morphological predictions, but the hit count has saturated. This is consistent with the tokenizer-bound nature of MorphAcc: once the model learns the morpheme boundaries that the 4K tokenizer makes available, further training cannot improve MorphAcc without a better tokenizer (Section 7.2).

Context matters: Bare nouns (`etxe`, `mendi`) produce probability distributions dominated by punctuation and fragments — the model needs syntactic context to predict case suffixes. With full sentence context (`Bihar...nire etxe -> ra`), the model correctly places significant probability mass on the correct suffix. This confirms the Mamba-2 architecture *can* learn morphology — it just needs sufficient context and training.

Comparison to old 32K-vocab model: The old 32K-vocab model at step 30K achieved only 20% MorphAcc (1/5 tests). The 4K-vocab model’s 70-76% MorphAcc represents a dramatic improvement, validating the vocabulary-size ablation (Section 4.5.4).

6.5 Case Paradigm Completion

Results (84 tests: 6 roots x 14 cases, f16 GPU):

Metric	Step 32K	Step 54K	Step 74K
Paradigm Hit@1	13.1% (11/84)	13.1% (11/84)	10.7% (9/84)
Paradigm Hit@3	20.2% (17/84)	21.4% (18/84)	22.6% (19/84)
Paradigm Hit@5	28.6% (24/84)	27.4% (23/84)	27.4% (23/84)

The paradigm metrics are noisy across checkpoints — Hit@1 actually *drops* from 13.1% to 10.7% between 54K and 74K, while Hit@3 *improves* from 21.4% to 22.6%. Hit@5 is flat. This noise further confirms that the autocomplete-specific metrics do not reliably track model quality: the paradigm test is high-variance (84 tests, bare-root prompts) and small improvements in PPL do not produce monotonic improvements in case suffix prediction. The absolute case (-a, the citation form) is

well-learned (83% Hit@5 at all checkpoints). The ergative (-ak) and inessive (-an) show partial learning. Most other cases remain below threshold — morphology emerges late and requires more context than bare-root prompts provide.

6.6 Blinded Human A/B Evaluation

Methodology: 30 fresh prompts from held-out validation text (zero overlap with CSR held-out set). Both checkpoints generate greedy completions (15 tokens max, f16, reference sentencepiece, no BOS, no quantization confound). A/B randomly assigned per prompt (different mapping in each batch). The Basque expert judges quality without knowing which checkpoint is A or B.

Results:

	Step 32K	Step 54K	Ties
Batch 1 (20 prompts)	7	6	7
Batch 2 (10 prompts)	4	3	3
Combined (30 prompts)	11 (55%)	9 (45%)	10 (33%)

Binomial test (two-sided): $p = 0.82$ — definitive statistical tie. An 11-vs-9 split out of 20 decisive judgments happens ~41% of the time by pure chance. The two checkpoints are statistically equivalent in autocomplete quality.

Expert qualitative observations: 1. **Repetition loops** — both models fall into them on some prompts (greedy-decoding failure mode, not checkpoint-specific) 2. **“Both could be correct”** — the expert noted this on ~10/30 prompts, confirming the agglutinative multiple-valid-forms problem that caps CSR 3. **“Hard to judge without the full sentence”** — autocomplete quality is context-dependent and genuinely ambiguous

6.7 Cross-Model Baseline Comparison

To contextualize Morpheus’s performance, we evaluated two external Basque language models under the same evaluation protocol: **HiTZ/gpt2-eus-euscrawl** (GPT-2 small, 124M parameters, trained on the EusCrawl corpus, ~423M tokens) and **HiTZ/Latxa-Qwen3.5-2B** (Latxa, 1.88B parameters, a Qwen3.5-based instruct model adapted for Basque, trained on the publicly available Latxa Corpus v2, ~4.2B tokens; Sainz et al., 2025). All three models were evaluated on the same corpus (`eval/real_corpus/`, 475,750 characters across 14 Wikipedia and Berria news files) and the same 30-sentence CSR test set.

BPC: The Correct Cross-Model Metric Per-token perplexity (PPL) is tokenizer-dependent and cannot be compared across models with different vocabulary sizes. A model with a 4K vocabulary produces more, shorter tokens than one with 50K, inflating per-token PPL without reflecting actual prediction quality. **Bits Per Character (BPC)** normalizes this: it computes the total negative log-likelihood in bits and divides by the number of characters, making it independent of tokenization choices.

Model	Params	Vocab	BPC	PPL (token)	Tok/Char
GPT-2 eus-euscrawl	124M	50K	0.981	29.21	0.202
Morpheus v2 (Mamba-2)	91M	4K	0.970	9.83	0.294
Latxa-Qwen3.5-2B	1,882M	248K	0.822	4.89	0.359

WARNING: Corpus contamination caveat: Wikipedia and Berria articles may appear in all three models’ training data. Absolute BPC values are optimistic; the relative comparison is valid.

Note on shared corpus: Morpheus and Latxa-Qwen3.5-2B both derive their training data from the Latxa Corpus v2 (Section 4.2). Morpheus uses a curated subset (11 of 14 sub-corpora, with additional deep-cleaning) trained for ~2.16 epochs (~10B tokens seen); Latxa-Qwen3.5-2B uses the same corpus for continued pretraining of a Qwen3.5 base (~4.2B tokens). The BPC difference between them is therefore attributable to model size (91M vs 1,882M), architecture (Mamba-2 vs Transformer), and training regime (from-scratch vs instruct-tuned continued pretraining) — not to data source differences. GPT-2 eus-euscrawl, by contrast, was trained on EusCrawl only (~423M tokens), making the Morpheus–GPT-2 comparison a natural experiment in data volume (24x difference) at similar parameter scale.

Key findings:

1. **Morpheus outperforms GPT-2 on BPC** (0.970 vs 0.981) despite having 27% fewer parameters (91M vs 124M). The primary driver is training data volume: Morpheus was trained on ~10B tokens vs GPT-2’s ~423M tokens — a 24x difference. This confirms that for low-resource languages, training data quantity matters more than parameter count. However, the near-tie in BPC despite the 24x data difference also reveals that character-level metrics saturate before morphological competence is achieved — see Section 6.8 for a detailed data scaling analysis.
2. **Latxa achieves the lowest BPC** (0.822), as expected for a 1.88B-parameter model — 20x larger than Morpheus. However, the BPC gap (0.148 bits/char) is modest relative to the parameter difference, reflecting diminishing returns from scale.
3. **PPL is misleading across vocabularies.** GPT-2’s per-token PPL of 29.21 appears catastrophic compared to Morpheus’s 9.83, but BPC reveals the models are nearly equivalent (0.981 vs 0.970). The apparent PPL gap is almost entirely an artifact of GPT-2’s 50K vocabulary producing longer tokens that are individually harder to predict.

Simplified Next-Word CSR (No Inference Engineering) To provide a fair raw-model comparison without our inference engineering advantages, we evaluated all three models with a simplified next-word CSR protocol: greedy decode until a word boundary, extract the first word, and compare to the target word.

Model	CSR (macro)	95% CI	Word Accuracy
GPT-2 eus-euscrawl	0.110	[0.060, 0.167]	37.6% (56/149)
Morpheus v2	0.094	[0.058, 0.131]	60.4% (90/149)
Latxa-Qwen3.5-2B	0.237	[0.201, 0.275]	68.5% (102/149)

Key findings:

1. **Latxa has the highest CSR** (0.237) and word accuracy (68.5%), consistent with its superior BPC. Its CIs do not overlap with Morpheus or GPT-2.
2. **GPT-2 and Morpheus have overlapping CIs** — the CSR difference (0.110 vs 0.094) is not statistically significant. However, **Morpheus has 1.6x higher word accuracy** (60.4% vs 37.6%), meaning it predicts the correct word far more often. This is another instance of the CSR paradox (Section 6.13): Morpheus predicts correct words more frequently but saves fewer keystrokes per correct prediction, because agglutinative Basque words are longer and require more characters before the model converges.
3. **Latxa produces noisy completions.** As an instruct model, Latxa generates web navigation artifacts, markdown headers, and repeated text when used for raw text completion (e.g., the prompt “Euskal Herriko” produces “bidaia-gida/Ibilbideak/Arriurdin mendia”). This highlights a key architectural advantage of Morpheus: as a **base language model** (not instruction-tuned), it is naturally suited for the autocomplete task, where the model must continue text seamlessly rather than follow instructions.
4. **Inference engineering adds 3.9x CSR.** Morpheus’s simplified CSR of 0.094 increases to 0.362 with the full inference pipeline (retokenization fallback, sticky merge, top-k alternatives — see Section 5.5). This demonstrates that the engineering strategies documented in this paper are not marginal optimizations but a major contribution, nearly quadrupling the raw model’s autocomplete utility.

6.8 Data Scaling Analysis: Training Data Requirements for Low-Resource Language Models

A natural question for any language model training effort is whether the training data budget was appropriately sized. We analyze this through the lens of established scaling laws and our empirical training trajectory.

Tokens-to-parameters ratio. Our model has 91M parameters (including embeddings) and was trained on ~10B tokens, yielding a ratio of approximately **110:1** (tokens per parameter). The relevant scaling law literature provides context:

Framework	Ratio	Source
Chinchilla (compute-optimal)	20:1	Hoffmann et al. (2022)
DeepSeek (quality-adjusted)	30:1	Bi et al. (2024)
Mosaic (inference-optimal)	190:1	Sardana et al. (2024)
MiniCPM (small-model optimal)	192:1	Hu et al. (2024)
Morpheus (actual)	110:1	—

Chinchilla’s 20:1 ratio is **compute-optimal** — it minimizes training compute for a target loss, but does not account for inference cost. Sardana et al. (2024) showed that for models with significant inference demand (~1B requests), the optimal ratio shifts upward: train smaller models on more data than Chinchilla recommends. Since our model is deployed as a per-keystroke autocomplete system (exactly the inference-heavy scenario Mosaic addresses), the Mosaic/M iniCPM range is more applicable than Chinchilla. Hu et al. (2024) specifically studied small language models (0.04B–2B parameters) and found a much higher optimal ratio (192:1) than Chinchilla, contradicting the intuition that small models need less data.

By modern small-model standards, our 110:1 ratio is not excessive. It sits below both the Mosaic inference-optimal (190:1) and MiniCPM small-model-optimal (192:1) recommendations. Modern small language models are trained at far higher ratios: Qwen3-0.6B at 60,000:1 (36T tokens), Liquid LFM2.5-350M at 80,000:1 (28T tokens).

Empirical convergence analysis. While the ratio is reasonable by scaling-law standards, our training trajectory reveals where the model actually stopped learning:

Training segment	Tokens seen	Delta	
		PPL	Tokens per 1.0 PPL improvement
Step 32K -> 54K	4.2B -> 7.1B	-0.39	7.4B
Step 54K -> 74K	7.1B -> 9.7B	-0.04	137.6B

The improvement rate dropped by **18.6x** between segments. The model effectively converged around **step 67K (~8.8B tokens)** — the final ~1.2B tokens produced no measurable PPL improvement. This does not mean the data budget was excessive; it means the **model capacity saturated on this data distribution**.

Data quality as the binding constraint. The convergence at 8.8B tokens is better explained by data quality than by data quantity. Three pieces of evidence converge:

1. **Corpus noise.** Our corpus quality audit (Section 4.2, Section 6.11) identified ~20–30% visible artifacts: emoji-heavy social media residue (18% of sampled lines), exact duplicates (5.2%), mixed-language content (6.4%), templated repetition, and date/number patterns. If ~20–30% of the corpus is low-value noise, the effective high-quality data is ~7–8B tokens — closely matching the 8.8B convergence point.
2. **GPT-2 eus-euscrawl as a natural experiment.** GPT-2 eus-euscrawl was trained on only ~423M tokens (3.4:1 ratio, heavily undertrained by Chinchilla standards) yet achieves nearly the same BPC as Morpheus (0.981 vs 0.970). However, Morpheus achieves **1.6x higher word accuracy** (60.4% vs 37.6%) — the 24x data advantage produces a dramatically better autocomplete model. This confirms that token count matters, but the BPC near-tie with a model trained on 24x less data suggests that character-level metrics saturate well before the model has learned the morphological patterns needed for good autocomplete. Quality multiplies the effectiveness of quantity.
3. **DeepSeek’s quality finding.** Bi et al. (2024) found that “the data quality significantly influences the optimal model/data scaling up allocation strategy” — higher-quality data means the compute budget should shift toward model size rather than data size, implying that high-quality data is more efficient per token.

Implication: data quality over quantity. A 2–5B token corpus of aggressively filtered, high-quality Basque text would likely match or exceed the current 10B token mixed-quality corpus, because the marginal ~1.2B tokens that produced no improvement and the ~2–3B tokens of noise were not contributing to model quality. The Chinchilla-optimal point for our model (~1.8B tokens) serves as a floor; the MiniCPM-optimal point (~17.5B tokens) is unattainable without more high-quality Basque data than currently exists. The practical sweet spot for low-resource agglutinative language modeling at this scale is **2–5B tokens of quality-filtered data**.

Smart Compose as a high-resource reference point. Google’s Smart Compose (Section 2.1, Table 1) provides a concrete reference: the production ~80M LSTM was trained on ~8 billion English emails (~320B+ tokens estimated) — roughly **30–60x our 10B token corpus**. Beyond volume, Smart Compose’s data is **domain-matched** (trained on emails, deployed for email writing), which is itself a form of data quality that Morpheus lacks (Section 6.11). This enormous data advantage is available because English is high-resource and Google had access to 1.5 billion users’ emails. For Basque, no comparable data volume or domain-matched corpus exists. This is the fundamental asymmetry: Google can scale data indefinitely; we cannot. The question is not “how much data should we use?” but “given that Basque data is finite and domain-matched corpora do not exist, how do we maximize quality from what exists?” The answer is aggressive quality filtering rather than raw quantity maximization.

This finding has implications for low-resource language modeling broadly: for languages where the total available high-quality text is finite and modest, data quality is the binding constraint, not data quantity.

6.9 Evaluation Reliability: PPL vs. Autocomplete Metrics

Signal	Paradigm	Favors 74K?	Significant?	What it measures
PPL (1.83M held-out tokens)	—	[OK] Yes (7.56->7.17->7.13)	[OK] Yes (all 14 files agree)	Language model quality (full distribution)
CSR (300 held-out sentences)	Multi-token	Directionally (24.90->25.23->25.26)	[X] CIs overlap	Keystroke economy (lower bound)
Human A/B (30 blinded)	Multi-token	[X] No (11 vs 9 for 32K)	[X] p=0.82	Autocomplete quality (ground truth)
Completion replay (real usage)	Next-word	[OK] Yes (Top-1 60->80%)	—	Chip acceptance hit rate
Typing simulation (15 sentences)	Next-word	—	—	Word accuracy, CSR paradox

Core finding: PPL is the only metric that produces coherent, trustworthy signal. It unambiguously confirms step 74K is the best model (7.56 -> 7.17 -> 7.13, all 14 files agree monotonically). The autocomplete metrics proved unreliable for checkpoint ranking:

- **CSR** is fragile to implement (string prompts gave ~4% vs ~28% with token-ID prompts due to BOS and tokenizer divergence bugs) and, even when correct, produces overlapping CIs at n=300. It uses exact-match gold and cannot credit valid alternative Basque continuations — it detects *regression* but not *progress* between competent models. This was confirmed empirically: a 30-sentence eval showed 54K as *worse* than 32K (noise); at n=300, the direction flipped to agree with PPL.
- **Human A/B** at n=30 is underpowered (p=0.82). A 60/40 split yields p=0.30; even 70/30 yields p=0.13. The expert confirmed ~33% ties due to multiple valid continuations.
- **MorphAcc** saturates at 76% from step 54K (tokenizer-bound).
- **Completion replay** favors 54K (Top-1 60->80%) but conflates model quality with inference engineering.

CSR is a lower bound — and structurally biased. The typing simulation (Section 6.13) reveals a deeper problem: CSR is not merely resolution-limited, it is **structurally biased against the target language**. Agglutinative morphology requires longer typed prefixes before prediction, consuming the keystroke savings CSR measures. The CSR inversion (Section 6.14) — where NW-CSR actually *decreases* as PPL improves — provides the strongest evidence that autocomplete metrics can actively mislead checkpoint selection. Notably, the expert’s **unblinded** impression that 54K was “much better” was not borne out by blinded A/B evaluation (statistical tie) — **blinded evaluation is essential for honest model comparison**.

6.10 Overall Assessment at Step 74K (Training Complete)

Metric	Result	Interpretation
Held-out PPL	7.13	Strong LM quality; converged (flat from step 67K)
CSR (macro, n=300)	25.26% [24.00, 26.52]	Meaningful keystroke savings; saturated
MorphAcc	76%	Strong morphological competence; saturated at 54K (tokenizer-bound)
Paradigm Hit@5	27.4%	Partial case system; noisy across checkpoints
Human A/B	Tie (p=0.82)	Statistical tie at n=30 (underpowered)
Q4_K_M size	55 MB	Well within 300 MB budget

The model is fully converged. The 54K->74K improvement is visible only in PPL (7.17 -> 7.13): CSR barely moves, MorphAcc is flat, Paradigm is noisy. Once a model reaches competence, only PPL has the resolution to measure further improvement.

6.11 Corpus-Induced Prediction Artifacts

A persistent quality issue observed during qualitative testing is the model’s tendency to autocomplete with **dates, numbers, and temporal expressions** in contexts where a human would predict more general continuations. For example:

Prompt	Model suggestion	Expected style
Aipatu bezala,	2015eko ekainean,	General continuation

The model predicts 2015eko ekainean (“in June 2015”) rather than a more broadly useful continuation. This is not a random error but a **systematic corpus bias**: the training corpus is dominated by encyclopedic (Wikipedia) and journalistic (Berria) prose, where sentences following phrases like “as mentioned,” “as stated,” or “according to” overwhelmingly reference specific dates, years, and quantities. The model has faithfully learned this distribution.

This artifact persists despite the exclusion of official gazette sources (BOG, BOPV, BOTHA), which were removed precisely because they contained bare numbers in sentence position (decree IDs, budget amounts). The residual date/number bias comes from **legitimate prose** — Wikipedia articles and news articles are inherently rich in temporal and numeric references. This represents a fundamental tension: the same encyclopedic and journalistic sources that provide high-quality, well-formed Basque prose also teach the model that dates and numbers are highly predictable continuations. It is also a manifestation of **domain mismatch**: the model is trained on encyclopedic and journalistic prose but deployed as a general-purpose text editor autocomplete. Google’s Smart Compose avoids this problem entirely because its training data (emails) matches its deployment context (writing emails) — the model learns exactly the distribution users produce (Section 6.8). For Basque, no large conversational or email corpus exists, so we train on the best available general prose and accept the resulting domain artifacts as a known limitation.

Implications for evaluation: This artifact is not captured by PPL (predicting frequent date patterns *lowers* PPL) and is only partially captured by CSR (the held-out sentences may or may not contain dates). It is most visible in open-ended autocomplete testing with real user prompts, where the mismatch between the model’s learned distribution and the user’s intent becomes apparent. This reinforces the finding that **no single metric is sufficient** and that qualitative testing with domain experts remains essential — a conclusion independently reached by the GitHub Copilot team, who report that “language-specific evaluations lead to better outcomes along quality and style preferences” beyond what execution-based testing, LLM-based evaluations, and A/B testing provide (Fu & Mogensen, 2026).

Mitigation directions are discussed in Section 7.1 (data-level) and Section 7.3 (domain fine-tuning).

6.12 Cross-Lingual Transfer

Although the model is trained exclusively on Basque-focused corpora, web-crawled sources and parliamentary transcripts inevitably contain small amounts of non-Basque text (~0.6% English by weighted volume; Spanish similarly present). We tested next-word prediction on common collocations in all three languages using the keyboard-mode demo endpoint. The model is **strongly Basque-specialized**: Basque prompts achieve 60% top-1 and 80% top-3 accuracy with 51% average confidence — roughly 3x the top-1 accuracy and 1.7x the confidence of English or Spanish. The model resolves ergative alignment contrasts correctly (*dugu* after a transitive frame, *gara* after an intransitive frame) and handles suffix attachment (*arreta* -> *arretagatik*, *Aldez* -> *aurretik*). Despite this Basque dominance, the model exhibits **weak incidental cross-lingual transfer** — correctly predicting high-frequency English/Spanish collocations (*Thank you very* -> *much*, *Los Estados* -> *Unidos*) but with low confidence and failing on less formulaic phrases. This is an **artifact of corpus composition, not a feature**: even aggressively monolingual Basque corpora contain enough multilingual contamination to produce measurable cross-lingual effects. The typing simulation (Section 6.13) confirms this transfer is functional, not merely collocational: the model sustains full 10–12 word English/Spanish sentences with 100% word accuracy.

Language	N prompts	Top-1	Top-3	Avg. max confidence
Basque	10	60.0%	80.0%	0.511
English	30	23.3%	23.3%	0.196
Spanish	30	16.7%	30.0%	0.299

6.13 The CSR Paradox: Agglutinative Morphology Penalizes Native-Language Keystroke Savings

To evaluate the keyboard-mode autocomplete under realistic usage, we developed a typing simulation that **faithfully replicates the frontend algorithm** (Section 5.5): char-by-char typing, sticky merge, top-3 chip display from a top-5 fetch, and full acceptance semantics. Fifteen sentences — five each in Basque, English, and Spanish — are translations of the same semantic content, controlling for topic and structure. The model accepts a suggestion the moment a matching candidate appears in the top-3 chips.

Results:

Language	Words correct	Simulated CSR	Avg. confidence	Acceptance rate	Avg. prefix before accept
Basque (native)	39/39 (100%)	19.2%	0.204	77%	4.7 chars
English (<1% corpus)	50/50 (100%)	26.3%	0.463	68%	3.1 chars
Spanish (<1% corpus)	46/46 (100%)	30.9%	0.590	70%	3.1 chars
Overall	135/135 (100%)	25.4%	0.424	71%	3.6 chars

The model’s native language achieves the lowest simulated CSR — below two languages that represent less than 1% of the training corpus. Spanish, the least-represented language, achieves the highest CSR. This is counterintuitive: one would expect the model to perform best on the language it was trained on.

This is **not a model deficiency**. It is a structural property of agglutinative morphology, and it reveals a fundamental flaw in using CSR as a primary metric for agglutinative autocomplete.

Root cause: morphological length. Basque words are longer and morphologically complex. The model needs an average of 4.7 typed characters before the correct Basque suggestion appears, versus 3.1 for English and Spanish. A word like *paseatzera* (“to go for a walk”) cannot be predicted from *pa* — the model must see *paseatzer* before it confidently offers the full word. In contrast, English *walk* is predictable from *w* once the collocational context (*go for a...*) is established. The keystroke savings are consumed by the longer prefix the user must type before a suggestion becomes available, even though the model ultimately predicts the correct word.

Confidence inversely correlates with CSR. Basque has the lowest average confidence on accepted suggestions (0.204) yet the highest acceptance rate (77%). The model is “cautiously correct” on Basque — it identifies the right word but with distributed probability mass across morphological variants (e.g., *paseatzera*, *paseatzeko*, *paseatzen* are all valid continuations of *paseatze*). English and Spanish, with shorter words and stronger collocational patterns, produce high-confidence predictions (English *language* at 0.993, Spanish *trabajando* at 0.991) but lower acceptance rates — many short function words (*the*, *is*, *a* / *el*, *y*, *un*) are too ambiguous at 1–2 characters to predict, dragging down the acceptance rate without reflecting on model quality.

Only **one high-confidence (≥ 0.8) acceptance occurred in Basque** (*zait*, 0.932), compared to 11 in English and 11 in Spanish. Yet Basque achieved 100% word accuracy — the model was always correct, just less certain. This is the signature of agglutinative prediction: multiple valid morphological continuations distribute probability mass, producing lower per-word confidence without indicating poorer predictions.

Implications for evaluation methodology. CSR is not merely a lower bound (Section 6.9) — it is a **structurally biased metric** that penalizes the very language the system is designed to serve. If CSR were used as a primary optimization target, it would systematically favor shorter-word languages. This aligns with GitHub’s Copilot experience, where acceptance-rate optimization “could lead to incorrectly favoring a high volume of simple and short suggestions” (Fu & Mogensen, 2026) — the same structural bias at production scale in a different domain.

Recommendation: CSR for agglutinative autocomplete should be accompanied by (1) word accuracy, (2) acceptance rate, and (3) average prefix length before acceptance. A naive reader comparing our 25.4% simulated CSR to the ~80% achievable in English autocomplete would conclude the model is poor, when in fact it achieves 100% word accuracy and 77% acceptance on its target language. The gap is structural, not qualitative.

6.14 The CSR Inversion: Autocomplete Metrics Move Opposite to Model Quality

The CSR paradox (Section 6.13) shows that CSR penalizes agglutinative languages structurally. A more troubling question is whether CSR tracks model quality *within* a single language across training. To test this, we ran the next-word CSR simulation (the PyTorch port of the keyboard algorithm, Section 5.5.6) on **seven checkpoints** spanning the full training trajectory, from step 10K (early training) through step 76K (converged). The same 30 Basque CSR test sentences were used at every checkpoint, and the simulation faithfully replicates the deployed keyboard algorithm (retokenization fallback, sticky merge, top-3 display, acceptance semantics).

Results across the full training trajectory:

Step	Held-out PPL	NW- CSR	95% CI	Word Acc	Acceptance	Avg Prefix	Confidence	Manual
10K	—	0.375	[0.313, 0.438]	1.000	0.859	3.5	0.408	14.1%
20K	—	0.402	[0.344, 0.469]	1.000	0.852	3.2	0.415	14.8%
30K	—	0.382	[0.330, 0.443]	1.000	0.839	3.4	0.422	16.1%
32K	7.56	0.397	[0.352, 0.450]	1.000	0.866	3.6	0.400	13.4%

Step	Held-out PPL	NW- CSR	95% CI	Word Acc	Acceptance	Avg Prefix	Confidence	Manual
54K	7.17	0.385	[0.355, 0.448]	1.000	0.859	3.6	0.415	14.1%
74K	7.13	0.362	[0.313, 0.426]	1.000	0.832	3.5	0.433	16.8%
76K	7.13	0.373	[0.320, 0.450]	1.000	0.832	3.3	0.432	16.8%

What is clear: the model improved (PPL 7.56 -> 7.13). **What is unexpected:** NW-CSR does not track this. It peaks at step 20K (0.402) and reaches its lowest at step 74K (0.362) — the best checkpoint. If CSR had been the selection criterion, it would have chosen step 20K over step 74K. We call this the **CSR inversion**: the autocomplete metric moves opposite to model quality.

The decomposed metrics reveal: confidence increases monotonically (0.408 -> 0.433) while acceptance decreases (0.859 -> 0.832). Word accuracy remains perfect (1.000) at every checkpoint. The decline is not quality degradation.

Hypotheses (preliminary, unresolved). We do not yet understand why this occurs and cannot rule out computation artifacts: 1. **Multiple valid continuations** — better model converges on *a* valid answer, not *the* gold answer; exact-match CSR cannot credit it. 2. **Agglutinative probability distribution** — better model distributes probability across morphological variants, depressing top-1 on any single one. 3. **Computation artifacts** — retokenization fallback merges probabilities across non-comparable paths; sticky merge +0.1 boost may interact with checkpoint-specific rankings. 4. **Evaluation set sensitivity** — 30 sentences from a single Wikipedia topic; stylistic drift as training progresses. 5. **Small sample size** — all CIs overlap (n=30); the trend is directional, not proven.

Practical implication: CSR should not be used as a primary checkpoint selection criterion (Section 6.9). The inversion shows CSR can actively *prefer* a worse model.

GGUF cross-validation: the inversion is backend-dependent. We ran the same 30 sentences through the deployed GGUF model (Q5_K_M, llama.cpp). The inversion **does not replicate in GGUF**: PyTorch CSR decreases monotonically (0.397 -> 0.362) while GGUF is roughly flat (0.362 -> 0.389), with step 32K *lowest* in GGUF despite being *highest* in PyTorch — supporting hypothesis (3) that bf16 forward pass vs. Q5_K_M dequantization produce different logprob distributions. Full cross-backend comparison table in Appendix D.

Both backends agree: neither shows CSR increasing with model quality — the claim that CSR should not be used for checkpoint selection holds regardless of backend, though the stronger claim that CSR *inverts* is backend-specific. Quantization also reduces confidence across all checkpoints (e.g., step 74K: 0.433 -> 0.381), as expected from 5-bit weight encoding.

7. Future Work

7.1 Immediate

1. **Training is complete** (76K steps, ~10B tokens, 14.9 hours). PPL flat from step 67K. Next: investigate the CSR inversion (Section 6.14) with a larger, more diverse evaluation set (100+

sentences).

2. **Investigate repetition loops** — both checkpoints exhibit greedy-decoding repetition on some prompts. Repetition penalty or nucleus sampling may improve practical autocomplete quality more than further training.
3. **Mitigate date/number prediction artifacts** (Section 6.11). The model over-predicts temporal and numeric expressions because the corpus is dominated by encyclopedic and journalistic prose. Candidate mitigations include: (a) **downweighting lines with high digit density** during training, (b) **post-hoc filtering** of numeric-heavy suggestions in the demo server, or (c) **domain reweighting** to reduce the relative proportion of Wikipedia/news text in favor of more conversational or instructional prose. Each approach has tradeoffs: downweighting may harm legitimate date prediction, post-hoc filtering adds latency, and domain reweighting requires additional clean data sources.
4. **Scale CSR evaluation** — 300 sentences with CIs is a significant improvement over 30, but 1000+ would further tighten confidence intervals.

7.2 Near-term (requires morphological analyzer)

1. **Integrate Apertium Basque as the morphological analyzer.** Apertium is the only practical Basque tool that emits explicit morpheme boundaries (+ symbols) rather than only lemmas or UD features. Throughput is $\sim 9.4 \times 10^5$ words/second, sufficient for a 22 GB corpus pass. The output must be **surface-preserving**: use segmented forms like `etxe#tik` rather than underlying morphology (`etxe+a+tik`) that doesn't match the visible text.
2. **Pre-segment corpus with surface-preserving morpheme boundaries** and retrain a morphology-aware tokenizer. Based on QuechuaTok results, this should increase MorphAcc from $\sim 67\%$ toward $\sim 80\%+$.
3. **Distill a mobile-variant model** for the next-word prediction paradigm (Section 5.4.1). The 91M model (55 MB) is sized for desktop (Smart Compose scale); mobile requires Gboard-scale (~ 5 -10M, ~ 5 -10 MB). A distilled Morpheus-Mobile using the 91M model as teacher would bring the inference engineering strategies (Section 5.5) into a smartphone-sized model. Mamba-2's $O(1)$ memory (no KV cache) is a key advantage for this target.

7.3 Medium-term

1. **Train Morpheus-Base (207M)** on the 4K tokenizer and compare against Morpheus-Small. A larger model may produce autocomplete improvements that are statistically detectable at current sample sizes — the 91M model's gains between 32K and 54K were directionally positive but too small for the multi-token metrics to confirm.
2. **MWE token injection**: Extract the 1,000 most frequent Basque multi-word expressions and inject them as single tokens. This directly reduces autoregressive decoding steps by 40-60% for covered phrases.
3. **Extend cross-model comparison** (Section 6.7): The current BPC comparison includes GPT-2 (124M), Latxa-Qwen3.5-2B (1.88B), and Morpheus (91M). A larger Latxa variant (7B+) and a base (non-instruct) Latxa model would provide a cleaner baseline, as the current Latxa's instruct-tuning produces noisy completions for raw text completion. Additionally, applying the full inference engineering pipeline (Section 5.5) to the baseline models would isolate the contribution of inference engineering from model quality.
4. **Domain-specific fine-tuning.** The base model is trained on general Basque prose (Wikipedia,

news, literature), which produces a general-purpose autocomplete but also corpus-induced artifacts such as the date/number prediction bias documented in Section 6.11. A lightweight fine-tuning stage on domain-specific corpora — e.g., legal Basque (terminology, statute language), medical, educational, or conversational/informal text — could improve suggestion relevance for specialized use cases while also mitigating the general-corpus artifacts. The current training infrastructure supports pretraining from scratch and checkpoint resume (continuing the same run), but **does not yet support fine-tuning-specific features**: separate fine-tuning datasets, layer freezing, differential learning rates, or parameter-efficient methods such as LoRA. The 91M parameter scale is small enough that full fine-tuning on a single GPU is feasible; the on-device deployment constraint means domain-adapted variants could be distributed as separate GGUF files and hot-swapped at runtime via the demo server’s model reload endpoint (Section 5.4). This is also a candidate mitigation for the Section 6.11 date/number artifacts: a domain adapter trained on conversational or instructional prose would shift the model’s distribution away from the encyclopedic patterns that produce numeric predictions.

5. **Data scaling experiment for Basque.** The analysis in Section 6.8 suggests that 2–5B tokens of aggressively filtered, high-quality Basque text would likely match or exceed the current 10B token mixed-quality corpus. To empirically determine the optimal data size, train four models with identical architecture on 1B, 2B, 5B, and 10B token subsets of a quality-filtered corpus, and compare held-out PPL at convergence. This would yield the first data scaling law curve for Basque — a contribution to low-resource language modeling methodology. The experiment would also test whether the convergence at ~8.8B tokens (Section 6.8) is a property of data quality (fixable with better filtering) or of model capacity (requiring a larger model to benefit from more data).

7.4 Long-term

1. **Hybrid Mamba-Attention architecture:** If pure Mamba-2 struggles with long-range morphological agreement, add 2-4 attention layers (“Jamba-style” hybrid).
2. **Distillation from Kimu 9B:** If a distilled 200M Transformer from the 9B Basque teacher becomes feasible, compare against the pure Mamba baseline.
3. **User study with CSR measurement:** Deploy to real Basque speakers and measure actual keystroke savings with A/B testing.

8. Code Availability

All code, model checkpoints, and evaluation artifacts are publicly available at github.com/itzune/morpheus-mamba. The repository contains:

- **Training pipeline:** Mamba-2 training with atomic checkpointing, gradient accumulation, and W&B integration
- **Data pipeline:** Corpus cleaning, SentencePiece Unigram tokenizer training, and pretokenization with validation leakage prevention
- **Export pipeline:** PyTorch -> HuggingFace -> GGUF conversion with BOS-token configuration
- **Evaluation suite:** PPL, CSR with bootstrap confidence intervals, MorphAcc, case paradigm completion, and blinded A/B evaluation

- **Demo server:** Docker-based FastAPI inference server supporting both multi-token ghost-text continuation (Smart Compose-style) and next-word prediction (smartphone keyboard-style chips), with completion logging and checkpoint replay
- **Tokenizer ablation data:** Full per-word tokenization results across 4K/8K/16K/32K vocabularies

The primary model checkpoint (step 74K, fully trained) and all quantized GGUF variants (Q4_K_M, Q5_K_M) are included. The model is also published on HuggingFace: `itzune/morpheus` (safetensors format) and `itzune/morpheus-gguf` (GGUF format).

9. Conclusion

Morpheus demonstrates that **State Space Models (Mamba-2) are a viable architecture for on-device predictive autocompletion in agglutinative languages**. The central research question — whether a Smart Compose-equivalent system can run **entirely on a consumer laptop** for text editors — is answered affirmatively: our 91M Mamba-2 model (55 MB, zero network calls) provides multi-token ghost-text autocompletion on-device, comparable in scale to Smart Compose’s ~80M but without the data-center dependency (Table 1, Section 2.1). After full training (76K steps, ~10B tokens, 14.9 hours), the best checkpoint (step 74K) achieves:

- **Held-out PPL of 7.13** (converged, flat from step 67K)
- **25.3% CSR** (95% CI [24.0%, 26.5%]) on 300 held-out sentences
- **76% MorphAcc** — 3.8x improvement over the 32K tokenizer’s 20%
- **55 MB on-disk** (Q4_K_M) — deployable on consumer laptops

Primary Contributions

1. **Vocabulary-size ablation for Basque.** Unigram MorphAcc drops from 66.7% at 4K to 28.6% at 32K, mirroring QuechuaTok. At 4K, verbal agreement morphemes (`zki`, `zu`, `ke`) are independently accessible; at 32K, they are buried in opaque atomic units. **Fertility is a confound, not a quality metric, for agglutinative tokenizer evaluation.**
2. **Evaluation methodology: PPL is the only reliable metric; autocomplete metrics are fragile.** PPL improved monotonically (7.56 -> 7.17 -> 7.13, all 14 files agree) and is the only metric that unambiguously ranked checkpoints. CSR is fragile to implement (string prompts gave ~4% vs ~28% with token IDs due to BOS/tokenizer bugs) and saturates at small sample sizes (CIs overlap at n=300). Human A/B at n=30 is underpowered (p=0.82). The **CSR paradox** (Section 6.13) shows the model’s native Basque achieves the *lowest* simulated CSR (19.2%) — below English/Spanish at <1% of training data — because agglutinative word length requires longer typed prefixes. The **CSR inversion** (Section 6.14) shows NW-CSR can actively *decrease* as PPL improves. **CSR penalizes the very language such systems are designed to serve and should not be used as a primary optimization target.** This is corroborated by GitHub’s Copilot team, who found acceptance-rate optimization structurally misleading (Fu & Mogensen, 2026).
3. **Three-gate pre-training validation protocol.** Corpus audit, proxy overfit test, and autocomplete smoke test that jointly validate data quality, pipeline integrity, and inference-time behavior before committing GPU resources.

4. **Inference engineering for agglutinative keyboards.** Five strategies — retokenization fallback, sticky merge, top-k exceeding display-k, next-word candidate extraction, and completion logging with replay — that address failure modes unique to subword-tokenized keyboards. Inference engineering adds 3.9x CSR on top of the raw model, demonstrating it is a major contribution, not a marginal optimization. We also report a critical dependency: Mamba-2 models require `llama.cpp >= commit dc2187d48` to avoid silently incorrect greedy outputs.
5. **Cross-model baseline comparison using BPC.** BPC is the correct tokenizer-independent metric. Morpheus (91M, BPC 0.970) matches GPT-2 eus-euscrawl (124M, BPC 0.981) — driven by 24x more training data, not architecture — and approaches Latxa-Qwen3.5-2B (1.88B, BPC 0.822) at 1/20th the parameters. Per-token PPL is misleading across vocabularies.
6. **On-device Smart Compose feasibility.** A Smart Compose-equivalent system can run entirely on-device (91M Mamba-2, 55 MB, ~10B tokens) for a morphologically complex language. Both Google and we chose recurrent architectures over Transformers to avoid KV-cache latency (Section 2.1). The data scale asymmetry (Google’s 30–60x larger corpus) reflects the high-resource vs. low-resource divide (Section 6.8).
7. **Data scaling analysis for low-resource LM.** Our 110:1 tokens-to-parameters ratio is reasonable by modern small-model standards (below Mosaic 190:1, MiniCPM 192:1). However, the model converged at ~8.8B tokens — not because the budget was excessive, but because **data quality is the binding constraint**. ~20–30% corpus noise means effective high-quality data is ~7–8B tokens, matching the convergence point. For low-resource languages, aggressive quality filtering of a 2–5B token corpus would likely outperform maximizing raw token count.

Limitations

- CSR of 25% is below the ~80% achievable in English autocomplete, reflecting both agglutinative difficulty and the **structural CSR paradox** (Section 6.13)
- **Date/number prediction artifacts** (Section 6.11): domain mismatch between training corpus (encyclopedic/journalistic) and deployment context (general text editor)
- **Evaluation metric limitations** (Section 6.9): PPL is the only reliable checkpoint-ranking metric; CSR, MorphAcc, and A/B all saturate or are underpowered
- No morphological pre-segmentation (Apertium) yet; MorphAcc could improve from 76% toward 83%+
- No user study; all evaluation is simulation-based or expert-judged
- The 91M model is too large for mobile (Gboard: 1.4M/1.4MB defines the mobile target); a distilled variant (~5–10M) has not yet been trained
- Data quality is the binding constraint (Section 6.8); a 2–5B token high-quality subset would likely match current quality but has not been empirically tested

The Path Forward

The model has converged (PPL flat from step 67K). The next steps are: (1) integrate **Apertium Basque** for surface-preserving morpheme pre-segmentation, pushing MorphAcc toward 83%; (2) apply aggressive quality filtering to a 2–5B token subset; (3) distill a **Morpheus-Mobile** (~5–10M) for the Gboard paradigm; (4) investigate the CSR inversion with larger evaluation sets.

Appendix A: The Morfessor Attempt and Pivot

Our first approach to improving morphological alignment was **morphological pre-segmentation** — training a Morfessor 2.0 model on the corpus, using it to segment words into morphemes before tokenizer training, then training a Unigram tokenizer on the pre-segmented text. This followed v1’s ADR-002 (“Morfessor Pre-Segmentation as Mandatory Pre-Processing Step”), which had been accepted but never executed.

We trained Morfessor 2.0 on 4M words from a corpus sample. The segmentation quality was **poor** due to the mixed-language nature of the corpus (Basque + Spanish + English):

Input	Morfessor output	Problem
cookie	coo\ kie	English word segmented incorrectly
dutenez	duten\ ez	Should be dute\ nez or unsplit

Morfessor’s MDL objective fails on mixed-language text without language filtering: it cannot distinguish Basque morphology from arbitrary character sequences in foreign words. Language-filtering the corpus before Morfessor was deemed too costly for the expected benefit.

Pivot: Rather than fix Morfessor, we noticed that the QuechuaTok paper showed vocabulary size alone drives significant MorphAcc gains (Unigram 4K = 66.67% **without** any pre-segmentation). We pivoted to testing vocabulary size as the primary variable — a simpler experiment with a stronger literature basis. Morfessor pre-segmentation remains unexplored and is deferred to future work with Apertium (Section 7.2), which provides linguistically grounded morpheme boundaries rather than statistical guesses.

Appendix B: Where 4K Still Fails — Full Failure-Mode Table

The 4K tokenizer handles single-layer case suffixes well but struggles with two categories:

Word	4K tokenization	Expected	Problem
etxean	etxean	etxe\ an	Whole word not split (inessive)
etxearentzat	etxe a rentzat	etxe\ arentzat	Multi-layer suffix split incorrectly
etxearengatik	etxe aren gatik	etxe\ arengatik	Multi-layer suffix split incorrectly
lagunera	lagun era	lagun\ ra	Epenthetic e- fused with suffix
lagunetik	lagun etik	lagun\ tik	Epenthetic e- fused with suffix

Two failure modes: 1. **Multi-layer suffixes** (a+ren+tzat): when a case suffix itself decomposes into multiple morphemes, 4K sometimes splits at the wrong internal boundary. 2. **Epenthetic vowels:** Basque inserts e- before consonant-initial suffixes after certain roots (lagun + tik ->

lagunetik). The 4K tokenizer fuses this epenthetic vowel with the suffix (lagun + etik) rather than with the root.

These remaining failures motivate the Apertium pre-segmentation future work (Section 7.2): a morphological analyzer that provides explicit, linguistically grounded boundaries would resolve both multi-layer suffixes and epenthetic vowel placement — problems that vocabulary size alone cannot fully solve.

Appendix C: Inference Engine Sensitivity — The SSM_SCAN Bug

During development, step 54K appeared to have “forgotten” the word *Kaixo* (predicting it at 0% probability) while step 32K predicted it correctly at 47.5%. Investigation via the replay system revealed that the model was not at fault: a stale Docker cache had built an older version of `llama.cpp` that contained a bug in the SSM (State Space Model) scan computation for Mamba-2 architectures (`n_groups > 1`, fixed in commit `dc2187d48`). The bug produced silently incorrect greedy outputs for certain weight configurations — step 54K’s weights triggered it, step 32K’s did not.

Recommendation: When deploying Mamba-2 models with `llama.cpp`, pin to a build that includes commit `dc2187d48` (2025-07-04 or later). If a checkpoint appears to have regressed, rebuild the inference engine with `--no-cache` before investigating the model. The completion logging and replay system (Section 5.5.6) is invaluable for detecting such issues, as it enables direct A/B comparison of checkpoints on identical inputs.

Appendix D: CSR Inversion — Full GGUF Cross-Validation Table

Step	Backend	NW-CSR	Acceptance	Confidence	Manual
32K	PyTorch (bf16)	0.397	0.866	0.400	13.4%
32K	GGUF (Q5_K_M)	0.362	0.852	0.372	14.8%
54K	PyTorch (bf16)	0.385	0.859	0.415	14.1%
54K	GGUF (Q5_K_M)	0.390	0.839	0.413	16.1%
74K	PyTorch (bf16)	0.362	0.832	0.433	16.8%
74K	GGUF (Q5_K_M)	0.389	0.852	0.381	14.8%

The inversion does not replicate in GGUF: PyTorch CSR decreases monotonically (0.397 -> 0.362) while GGUF is roughly flat (0.362 -> 0.389), with step 32K *lowest* in GGUF despite being *highest* in PyTorch. This supports hypothesis (3): bf16 forward pass vs. Q5_K_M dequantization produce different logprob distributions, amplified by sticky merge and retokenization fallback. Word accuracy remains perfect (1.000) in both backends at all checkpoints.

References

- Contreras, M. (2026). *QuechuaTok: Morphological Boundary Accuracy as a Necessary Metric for Tokenizer Evaluation in Agglutinative Low-Resource Languages*. arXiv:2606.23943.

2. Arnett, C., Hudspeth, M., & O'Connor, B. (2025). *Evaluating Morphological Alignment of Tokenizers in 70 Languages*. arXiv.
3. Stephen, A., & Libovický, J. (2026). *Evaluating Morphological Plausibility of Subword Tokenization via Statistical Alignment with Morpho-Syntactic Features*. arXiv.
4. Xu, N., & Kim, A. (2026). *Tokenization and Morphological Fidelity in Uralic NLP: A Cross-Lingual Evaluation*. arXiv.
5. Táboas García, T., Przybyla, P., & Wanner, L. (2025). *Exploring morphology-aware tokenization: A case study on Spanish language modeling*. EMNLP 2025.
6. Hu, J. F. (2025). *Tokenization Strategies for Low-Resource Agglutinative Languages in Word2Vec: Case Study on Turkish and Finnish*. ACDSA 2025.
7. Etxaniz, J., Sainz, O., Perez, N., Aldabe, I., Rigau, G., Agirre, E., Ormazabal, A., Artetxe, M., & Soroa, A. (2024). *Latxa: An Open Language Model and Evaluation Suite for Basque*. arXiv:2403.20266.
8. Chen, M. X., et al. (2019). *Gmail Smart Compose: Real-Time Assisted Writing*. arXiv:1906.00080.
9. Hard, A., et al. (2018). *Federated Learning for Mobile Keyboard Prediction*. arXiv:1811.03604.
10. Fu, S., & Mogensen, J. (2026). *The Road to Better Completions: Building a Faster, Smarter GitHub Copilot with a New Custom Model*. GitHub Blog.
11. Cheney, D. (2025). *How GitHub Copilot Serves 400 Million Completion Requests a Day*. QCon San Francisco 2024 / InfoQ.
12. Gu, A., & Dao, T. (2023). *Mamba: Linear-Time Sequence Modeling with Selective State Spaces*. arXiv:2312.00752.
13. Dao, T., & Gu, A. (2024). *Transformers are SSMS: Generalized Models and Efficient Algorithms through Structured State Space Duality*. arXiv:2405.21060.
14. Beck, M., et al. (2024). *xLSTM: Extended Long Short-Term Memory*. NeurIPS 2024.
15. Lane, W., Harrigan, A., & Arppe, A. (2022). *Interactive Word Completion for Plains Cree*. ACL 2022.
16. Trnka, K., & McCoy, K. (2008). *Evaluating Word Prediction: Framing Keystroke Savings*. ACL 2008.
17. Liquid AI (2026). *LFM-2.5-230M: On-Device State Space Models*. liquid.ai.
18. Orai NLP (2026). *Kimu: Basque-Adapted Language Models*. orai.eus.
19. ChaI-TeA (2024). *ChaI-TeA: A Benchmark for Chinese Input Method*. arXiv:2412.18377.
20. Kosyak, A., & Tyers, F. (2022). *Predictive text for agglutinative languages*.
21. WSTypist (2026). *Simulation-based mobile typing evaluation*. arXiv:2602.06489.
22. Rust, P., et al. (2021). *How Good is Your Tokenizer? On the Monolingual Performance of Multilingual Language Models*. ACL 2021.
23. Hoffmann, J., et al. (2022). *Training Compute-Optimal Large Language Models (Chinchilla)*. arXiv:2203.15556.
24. Sardana, N., et al. (2024). *Beyond Chinchilla-Optimal: Accounting for Inference in Language Model Scaling Laws*. arXiv:2401.00448.
25. Hu, S., et al. (2024). *MiniCPM: Unveiling the Potential of Small Language Models with Scalable Training Strategies*. arXiv:2404.06395.
26. Bi, X., et al. (2024). *DeepSeek LLM: Scaling Open-Source Language Models with Longtermism*. arXiv:2401.02954.
27. Muennighoff, N., et al. (2023). *Scaling Data-Constrained Language Models*. arXiv:2305.16264.
28. Sainz, O., et al. (2025). *Instructing Large Language Models for Low-Resource Languages: A Systematic Study for Basque*. EMNLP 2025.

This document is a living research record. Current as of July 11, 2026 — training complete (76,294 steps, best checkpoint step 74,000, PPL 7.13).